Confidence Intervals & Hypothesis Testing

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Constructing a confidence interval**

A random sample of 50 students has a mean study time of $\bar{x} = 6.2$ hours per week, with sample standard deviation $s = 1.8$ hours. Construct a 95% confidence interval for the population mean study time, using a critical value of $z^* \approx 1.96$.

Standard error: $\frac{s}{\sqrt{n}} = \frac{1.8}{\sqrt{50}} \approx 0.255$. Margin of error: $1.96(0.255) \approx 0.499$. Confidence interval: 6.2 ± 0.499, or approximately $(5.70, 6.70)$ hours.

2 **[Foundational]** • **Target: Constructing a confidence interval**

Interpret the confidence interval from the previous problem in the context of the study time question, using correct statistical language.

We are 95% confident that the true mean weekly study time for the population of all students is between approximately 5.70 and 6.70 hours — meaning that if we repeated this sampling process many times, about 95% of the resulting confidence intervals would capture the true population mean.

3 **[Foundational]** • **Target: Margin of error and sample size**

Explain two ways to decrease the margin of error of a confidence interval, and state a tradeoff involved with each.

(1) Increase the sample size n: since margin of error is proportional to $\frac{1}{\sqrt{n}}$, a larger sample shrinks the margin of error, but larger samples cost more time and resources to collect. (2) Decrease the confidence level (e.g. from 95% to 90%): a lower confidence level uses a smaller critical value, shrinking the margin of error, but this means being less confident that the interval actually captures the true parameter.

4 **[Foundational]** • **Target: The logic of hypothesis testing**

State the null hypothesis H_0 and alternative hypothesis H_a for a test of whether a coin is biased toward heads (true proportion of heads p), and classify the test as one-tailed or two-tailed.

$H_0 : p = 0.5$ (the coin is fair). $H_a : p > 0.5$ (the coin is biased toward heads). This is a one-tailed (right-tailed) test, since the alternative specifies a direction.

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5 [Foundational] • Target: The logic of hypothesis testing

Explain what a p-value represents, in the context of a hypothesis test.

The p-value is the probability of obtaining a test statistic at least as extreme as the one observed, assuming the null hypothesis is true. A small p-value suggests the observed data would be unusual if H_0 were true, providing evidence against H_0.

6 [Foundational] • Target: Four-step hypothesis testing procedure

A manufacturer claims their light bulbs last on average $\mu = 1000$ hours. A sample of 40 bulbs has mean lifetime $\bar{x} = 985$ hours with sample standard deviation $s = 45$ hours. State the hypotheses and compute the test statistic for testing whether the true mean lifetime is less than claimed.

$H_0 : \mu = 1000$. $H_a : \mu < 1000$. Test statistic: $z = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{985 - 1000}{45/\sqrt{40}} = \frac{-15}{7.115} \approx -2.11$.

7 [Foundational] • Target: Four-step hypothesis testing procedure

Using the test statistic $z \approx -2.11$ from the previous problem and a p-value of approximately 0.017, state a conclusion at the $\alpha = 0.05$ significance level.

Since the p-value (0.017) is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence that the true mean bulb lifetime is less than the manufacturer's claimed 1000 hours.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Type I and Type II errors, and power

In the light bulb test above, describe what a Type I error would mean in context, and what a Type II error would mean in context.

A Type I error would mean concluding the true mean lifetime is less than 1000 hours when it actually is not (rejecting a true H_0) — wrongly penalizing the manufacturer. A Type II error would mean failing to detect that the true mean lifetime really is less than 1000 hours (failing to reject a false H_0) — missing a real problem with the bulbs.

9 [Intermediate] • Target: Type I and Type II errors, and power

Explain what the power of a test measures, and state one way to increase it.

Power measures the probability that a test correctly rejects H_0 when the alternative hypothesis is actually true (i.e. correctly detecting a real effect). Power can be increased by increasing the sample size, which reduces variability in the sampling distribution and makes real effects easier to detect.

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10 **[Intermediate]** • **Target: Confidence intervals for a proportion**

A poll of 400 randomly selected voters finds that 220 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.

$\hat{p} = \frac{220}{400} = 0.55$. Standard error: $\sqrt{\frac{0.55(0.45)}{400}} = \sqrt{\frac{0.2475}{400}} \approx 0.0249$. Margin of error: $1.96(0.0249) \approx 0.0488$. Confidence interval: 0.55 ± 0.0488, or approximately $(0.501, 0.599)$.

11 **[Intermediate]** • **Target: Determining sample size for a target margin of error**

A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a conservative planning value of $p = 0.5$ (which maximizes required sample size). Find the minimum sample size needed, using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with $z^* = 1.96$.

$n = \left(\frac{1.96}{0.03}\right)^2 (0.5)(0.5) = (65.33)^2(0.25) \approx 4268.4(0.25) \approx 1067.1$. Round up: $n = 1068$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

12 **[Intermediate]** • **Target: One-sample t-test for a mean**

A nutritionist claims a new granola bar has a mean of 200 calories. A sample of 15 bars has mean $\bar{x} = 206$ calories with sample standard deviation $s = 9$ calories. Since the population standard deviation is unknown and n is small, a t-test is used. Compute the test statistic $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$.

$t = \frac{206 - 200}{9/\sqrt{15}} = \frac{6}{2.324} \approx 2.58$.

13 **[Intermediate]** • **Target: One-sample t-test for a mean**

Explain what it means for a hypothesis test result to be "statistically significant," and why statistical significance does not always imply practical importance.

A result is statistically significant when the p-value is small enough (below the chosen α) to reject H_0, meaning the observed data would be unlikely if H_0 were true. However, with a very large sample size, even a tiny, practically unimportant difference can become statistically significant — so significance alone doesn't guarantee the effect is large or meaningful in real-world terms; the actual size of the effect should also be considered.

14 **[Intermediate]** • **Target: One-sample t-test for a mean**

A researcher wants to test whether a new fitness program changes the average resting heart rate of participants from the population baseline of $\mu = 72$ bpm. A sample of 36 participants after the program has mean resting heart rate $\bar{x} = 68.5$ bpm with sample standard deviation $s = 10.5$ bpm. State the hypotheses (choosing a two-sided alternative), compute the test statistic, and using a p-value of approximately 0.11, state a conclusion at $\alpha = 0.05$.

$H_0 : \mu = 72$. $H_a : \mu \neq 72$. Test statistic: $t = \frac{68.5 - 72}{10.5/\sqrt{36}} = \frac{-3.5}{1.75} = -2.0$. Since the p-value (0.11) is greater than $\alpha = 0.05$, we fail to reject H_0 — there is not enough evidence that the fitness program changes the average resting heart rate.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: One-sample t-test for a mean** • **Cross-Lesson: 2.01 Sampling Methods & Bias**

Explain the relationship between a confidence interval and a two-sided hypothesis test at the same confidence/significance level — specifically, describe how you could use a 95% confidence interval alone to decide whether to reject $H_0 : \mu = \mu_0$ at $\alpha = 0.05$, without separately computing a p-value.

A 95% confidence interval and a two-sided test at $\alpha = 0.05$ are directly linked: if the hypothesized value μ_0 falls outside the 95% confidence interval, then $H_0 : \mu = \mu_0$ would be rejected at $\alpha = 0.05$; if μ_0 falls inside the confidence interval, H_0 would fail to be rejected. This means a confidence interval can be used as a quick visual/numeric check for significance without needing to separately calculate a p-value.

16 **[Challenging]** • **Target: One-sample t-test for a mean** • **Cross-Lesson: 2.02 Experimental Design & Scope of Inference**

A city council is deciding whether to fund a new public transit line, and wants to survey residents to estimate the true proportion who would use it regularly. A preliminary small survey suggests around 35% support, but the council wants a formal survey with a margin of error no larger than 2.5% at 95% confidence. Using $n = \left(\frac{z^*}{m}\right)^2 p(1-p)$ with the preliminary estimate $p = 0.35$ and $z^* = 1.96$, find the required sample size, rounding appropriately.

$n = \left(\frac{1.96}{0.025}\right)^2 (0.35)(0.65) = (78.4)^2 (0.2275) \approx 6146.6(0.2275) \approx 1398.4$. Round up: $n = 1399$ residents (sample size must be rounded up to guarantee the target margin of error is achieved).

17 **[Challenging]** • **Target: One-sample t-test for a mean** • **Cross-Lesson: 3.01 Probability Rules & Conditional Probability**

A restaurant chain wants to test whether a new menu item's average preparation time is longer than the target of 8 minutes, since customers have complained about slow service. A random sample of 30 orders for the new item has a mean preparation time of $\bar{x} = 8.7$ minutes with sample standard deviation $s = 1.8$ minutes. State the hypotheses for this one-sided test, compute the test statistic, and using a p-value of approximately 0.019, state a conclusion at $\alpha = 0.05$ in the context of the restaurant's concern.

$H_0 : \mu = 8$. $H_a : \mu > 8$. Test statistic: $t = \frac{8.7 - 8}{1.8/\sqrt{30}} = \frac{0.7}{0.329} \approx 2.13$. Since the p-value (0.019) is less than $\alpha = 0.05$, we reject H_0. There is convincing statistical evidence that the true mean preparation time for the new menu item exceeds the 8-minute target, supporting the customers' complaints about slow service.

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18 **[Challenging] • Target: Confidence intervals for a proportion • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions**

An agricultural researcher is designing an experiment to study the yield of a new variety of wheat. The experiment investigates two factors: - Factor 1: Fertilizer formula, with 3 different types. - Factor 2: Watering frequency, with 4 different schedules. If the experiment uses a completely randomized design where each subject receives a combination of one fertilizer formula and one watering schedule, how many distinct treatment groups are there?

In an experiment with multiple factors, a treatment is a specific combination of levels of each factor. 1. Identify the number of levels for each factor: - Factor 1 (Fertilizer) has 3 levels. - Factor 2 (Watering) has 4 levels. 2. Apply the multiplication counting principle to find the total number of distinct treatment combinations: $\text{Number of treatments} = 3 \times 4 = 12$ - Choosing 7 is a common misconception resulting from adding the levels of the factors ($3 + 4$) rather than multiplying them. - Choosing 64 results from inappropriately raising one number of levels to the power of the other (4^3). Final Answer: 12

19 **[Challenging] • Target: Determining sample size for a target margin of error • Cross-Lesson: 3.03 The Normal Distribution**

A study investigates the effectiveness of a new teaching method by comparing the test scores of students who used the new method versus those who used the traditional method. The results are summarized in a two-way table. If the table has 4 rows and 3 columns, what is the degrees of freedom (df) for a chi-square test of homogeneity?

For a chi-square test of homogeneity or independence with a two-way table, the degrees of freedom (df) are calculated using the formula: $\text{df} = (r-1)(c-1)$, where r is the number of rows and c is the number of columns. In this case, the number of rows is $r = 4$ and the number of columns is $c = 3$. Substituting these values into the formula: $\text{df} = (4-1)(3-1) = (3)(2) = 6$. Therefore, the degrees of freedom are 6.

20 **[Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem**

A researcher is conducting a survey on the relationship between college major and career satisfaction. They collected data from 150 students and organized it into a two-way table. If the number of rows in the table represents 3 different college majors and the number of columns represents 5 different levels of career satisfaction, what is the df for a chi-square test of independence?

For a chi-square test of independence, the degrees of freedom (df) are calculated using the formula $\text{df} = (r-1)(c-1)$, where r is the number of rows and c is the number of columns in the contingency table. In this case, $r = 3$ (college majors) and $c = 5$ (career satisfaction levels). Therefore, $\text{df} = (3-1)(5-1) = (2)(4) = 8$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Constructing a confidence interval

A sample of $n = 121$ has mean $\bar{x} = 393$. A claimed population mean is $\mu_0 = 409$, with sample standard deviation $s = 62$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 2.84

b. -2.34

c. -3.34

d. -2.84 ✓

$s/\sqrt{n} = 62/11 = 5.64$. $z = (393 - 409)/5.64 = (-16)/5.64 \approx -2.84$.

22 [Foundational] • Target: Margin of error and sample size

A sample of $n = 25$ has mean $\bar{x} = 513$. A claimed population mean is $\mu_0 = 495$, with sample standard deviation $s = 75$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -1.20

b. 0.70

c. 1.70

d. 1.20 ✓

$s/\sqrt{n} = 75/5 = 15.00$. $z = (513 - 495)/15.00 = (18)/15.00 \approx 1.20$.

23 [Foundational] • Target: The logic of hypothesis testing

A sample of $n = 25$ has mean $\bar{x} = 469$. A claimed population mean is $\mu_0 = 471$, with sample standard deviation $s = 51$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -0.20 ✓

b. 0.30

c. -0.70

d. 0.20

$s/\sqrt{n} = 51/5 = 10.20$. $z = (469 - 471)/10.20 = (-2)/10.20 \approx -0.20$.

24 [Foundational] • Target: Four-step hypothesis testing procedure

A sample of $n = 16$ has mean $\bar{x} = 157$. A claimed population mean is $\mu_0 = 183$, with sample standard deviation $s = 34$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -2.56

b. -3.06 ✓

c. 3.06

d. -3.56

$s/\sqrt{n} = 34/4 = 8.50$. $z = (157 - 183)/8.50 = (-26)/8.50 \approx -3.06$.

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25 [Foundational] • Target: Type I and Type II errors, and power

A sample of $n = 49$ has mean $\bar{x} = 187$. A claimed population mean is $\mu_0 = 162$, with sample standard deviation $s = 65$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 2.69 ✓

b. 3.19

c. -2.69

d. 2.19

$s/\sqrt{n} = 65/7 = 9.29$. $z = (187 - 162)/9.29 = (25)/9.29 \approx 2.69$.

26 [Foundational] • Target: Confidence intervals for a proportion

A sample of $n = 121$ has mean $\bar{x} = 348$. A claimed population mean is $\mu_0 = 346$, with sample standard deviation $s = 28$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. 1.29

b. 0.79 ✓

c. 0.29

d. -0.79

$s/\sqrt{n} = 28/11 = 2.55$. $z = (348 - 346)/2.55 = (2)/2.55 \approx 0.79$.

27 [Foundational] • Target: Determining sample size for a target margin of error

A sample of $n = 64$ has mean $\bar{x} = 241$. A claimed population mean is $\mu_0 = 244$, with sample standard deviation $s = 25$. Compute the test statistic $z = (\bar{x} - \mu_0)/(s/\sqrt{n})$, to two decimal places.

a. -0.46

b. -1.46

c. -0.96 ✓

d. 0.96

$s/\sqrt{n} = 25/8 = 3.12$. $z = (241 - 244)/3.12 = (-3)/3.12 \approx -0.96$.

28 [Foundational] • Target: One-sample t-test for a mean

A random sample of $n = 121$ has mean $\bar{x} = 180$ and sample standard deviation $s = 16$. Construct a 99% confidence interval for the population mean, using $z^* \approx 2.576$.

a. $(176.25, 181.75)$

b. $(175.25, 184.75)$

c. $(178.25, 183.75)$

d. $(176.25, 183.75)$ ✓

Standard error: $s/\sqrt{n} = 16/11 = 1.45$. Margin of error: $2.576(1.45) \approx 3.75$. Interval: 180 ± 3.75, or approximately $(176.25, 183.75)$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Constructing & Interpreting Confidence Intervals**

A random sample of $n = 49$ has mean $\bar{x} = 143$ and sample standard deviation $s = 41$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. $(130.52, 155.48)$

b. $(133.52, 154.48)$

c. $(131.52, 152.48)$

d. $(131.52, 154.48)$ ✓

Standard error: $s/\sqrt{n} = 41/7 = 5.86$. Margin of error: $1.96(5.86) \approx 11.48$. Interval: 143 ± 11.48, or approximately $(131.52, 154.48)$.

30 **[Intermediate]** • **Target: Hypothesis Testing Logic & P-values**

A random sample of $n = 121$ has mean $\bar{x} = 40$ and sample standard deviation $s = 11$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. $(37.04, 42.96)$

b. $(38.04, 41.96)$ ✓

c. $(38.04, 39.96)$

d. $(40.04, 41.96)$

Standard error: $s/\sqrt{n} = 11/11 = 1.00$. Margin of error: $1.96(1.00) \approx 1.96$. Interval: 40 ± 1.96, or approximately $(38.04, 41.96)$.

31 **[Intermediate]** • **Target: Type I Error, Type II Error, and Power**

A random sample of $n = 16$ has mean $\bar{x} = 108$ and sample standard deviation $s = 7$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. $(103.57, 112.43)$

b. $(106.57, 111.43)$

c. $(104.57, 111.43)$ ✓

d. $(104.57, 109.43)$

Standard error: $s/\sqrt{n} = 7/4 = 1.75$. Margin of error: $1.96(1.75) \approx 3.43$. Interval: 108 ± 3.43, or approximately $(104.57, 111.43)$.

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32 [Intermediate] • Target: One-Sample and Two-Sample t-Procedures

A random sample of $n = 49$ has mean $\bar{x} = 154$ and sample standard deviation $s = 59$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.

a. $(140.13, 167.87)$ ✓

b. $(139.13, 168.87)$

c. $(140.13, 165.87)$

d. $(142.13, 167.87)$

Standard error: $s/\sqrt{n} = 59/7 = 8.43$. Margin of error: $1.645(8.43) \approx 13.87$. Interval: 154 ± 13.87, or approximately $(140.13, 167.87)$.

33 [Intermediate] • Target: Constructing a confidence interval

A random sample of $n = 36$ has mean $\bar{x} = 23$ and sample standard deviation $s = 51$. Construct a 90% confidence interval for the population mean, using $z^* \approx 1.645$.

a. $(9.02, 34.98)$

b. $(9.02, 36.98)$ ✓

c. $(8.02, 37.98)$

d. $(11.02, 36.98)$

Standard error: $s/\sqrt{n} = 51/6 = 8.50$. Margin of error: $1.645(8.50) \approx 13.98$. Interval: 23 ± 13.98, or approximately $(9.02, 36.98)$.

34 [Intermediate] • Target: Margin of error and sample size

A random sample of $n = 16$ has mean $\bar{x} = 22$ and sample standard deviation $s = 56$. Construct a 95% confidence interval for the population mean, using $z^* \approx 1.96$.

a. $(-3.44, 49.44)$

b. $(-5.44, 49.44)$ ✓

c. $(-5.44, 47.44)$

d. $(-6.44, 50.44)$

Standard error: $s/\sqrt{n} = 56/4 = 14.00$. Margin of error: $1.96(14.00) \approx 27.44$. Interval: 22 ± 27.44, or approximately $(-5.44, 49.44)$.

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35 [Intermediate] • Target: The logic of hypothesis testing

A poll of 150 randomly selected voters finds that 37 support a candidate. Construct a 99% confidence interval for the true population proportion, using $z^* \approx 2.576$.

a. $(0.156, 0.307)$

b. $(0.136, 0.357)$

c. $(0.186, 0.337)$

d. $(0.156, 0.337)$ ✓

$\hat{p} = 37/150 = 0.247$. Standard error: $\sqrt{0.247(0.753)/150} \approx 0.035$. Margin of error: $2.576(0.035) \approx 0.091$. Interval: 0.247 ± 0.091, or approximately $(0.156, 0.337)$.

36 [Intermediate] • Target: Four-step hypothesis testing procedure

A poll of 300 randomly selected voters finds that 143 support a candidate. Construct a 95% confidence interval for the true population proportion, using $z^* \approx 1.96$.

a. $(0.450, 0.533)$

b. $(0.420, 0.533)$ ✓

c. $(0.400, 0.553)$

d. $(0.420, 0.503)$

$\hat{p} = 143/300 = 0.477$. Standard error: $\sqrt{0.477(0.523)/300} \approx 0.029$. Margin of error: $1.96(0.029) \approx 0.057$. Interval: 0.477 ± 0.057, or approximately $(0.420, 0.533)$.

37 [Intermediate] • Target: Type I and Type II errors, and power

A researcher wants a margin of error of at most 0.02 for a 90% confidence interval for a population proportion, using a planning value of $p = 0.70$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1-p)$ with $z^* = 1.645$.

a. 1421 ✓

b. 1420

c. 1371

d. 1471

$n = (1.645/0.02)^2(0.70)(0.30) \approx 1420.7$. Round up: $n = 1421$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Confidence intervals for a proportion • Cross-Lesson: 3.02 Discrete Random Variables & Binomial Distributions

A researcher wants a margin of error of at most 0.03 for a 95% confidence interval for a population proportion, using a planning value of $p = 0.40$. Find the minimum sample size needed, using $n = (z^*/m)^2 p(1-p)$ with $z^* = 1.96$.

a. 975

b. 1024

c. 1075

d. 1025 ✓

$n = (1.96/0.03)^2(0.40)(0.60) \approx 1024.4$. Round up: $n = 1025$ (sample size must always be rounded up to guarantee the margin of error requirement is met).

39 [Challenging] • Target: Determining sample size for a target margin of error • Cross-Lesson: 3.03 The Normal Distribution

A hypothesis test about a nutritionist's claimed mean calorie count produces a p-value of 0.112. Using a significance level of $\alpha = 0.01$, state the correct conclusion.

a. The test is inconclusive and must be repeated

b. **Fail to reject H_0; not enough evidence that the true mean calorie count differs from the claim** ✓

c. Reject H_0; the true mean calorie count differs from the claim

d. Accept H_0 as true

Since the p-value (0.112) is greater than $\alpha = 0.01$, we fail to reject H_0. Fail to reject H_0; not enough evidence that the true mean calorie count differs from the claim.

40 [Challenging] • Target: One-sample t-test for a mean • Cross-Lesson: 3.04 Sampling Distributions & Central Limit Theorem

A hypothesis test about a claimed mean reaction time produces a p-value of 0.034. Using a significance level of $\alpha = 0.05$, state the correct conclusion.

a. The test is inconclusive and must be repeated

b. Accept H_0 as true

c. Fail to reject H_0; not enough evidence that the true mean reaction time differs from the claim

d. **Reject H_0; the true mean reaction time differs from the claim** ✓

Since the p-value (0.034) is less than $\alpha = 0.05$, we reject H_0. Reject H_0; the true mean reaction time differs from the claim.

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41 [Challenging] • Target: Constructing & Interpreting Confidence Intervals • Cross-Lesson: 4.02 Chi-Square Tests & Regression Inference

What is the correct interpretation of a 95% confidence interval for a population mean $(42.1, 48.7)$?

a. We are 95% confident that the true population mean falls between 42.1 and 48.7 ✓

b. There is a 95% probability that the true mean is between 42.1 and 48.7

c. 95% of all individual values in the population lie between 42.1 and 48.7

d. 95% of sample means will equal the population mean

Confidence level refers to the long-run success rate of the method: in repeated sampling, 95% of intervals generated will capture the fixed true parameter.

42 [Challenging] • Target: Constructing & Interpreting Confidence Intervals • Cross-Lesson: 1.01 Measures of Center & Spread

How can you decrease the width of a confidence interval?

a. Increase the sample size or decrease the confidence level ✓

b. Increase the confidence level or increase the sample size

c. Decrease the sample size or increase the confidence level

d. Increase the standard deviation of the data

Width is $2 \times \text{critical value} \times \frac{\sigma}{\sqrt{n}}$. Increasing n shrinks the denominator; lowering confidence level (e.g. from 99% to 90%) lowers the critical value.

43 [Challenging] • Target: Hypothesis Testing Logic & P-values • Cross-Lesson: 1.02 Z-Scores & Linear Transformations

Why do statisticians state 'fail to reject H_0' instead of 'accept H_0'?

a. Failing to find sufficient evidence against H_0 does not prove that H_0 is true ✓

b. Because H_0 is always false in practice

c. To avoid computing the test statistic

d. Because the P-value is always approximate

Absence of evidence is not evidence of absence; we never prove the null hypothesis true, we merely conclude the data is consistent with it.

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44 [Challenging] • Target: Type I Error, Type II Error, and Power • Cross-Lesson: 1.03 Describing Distributions & Linear Regression

What is a Type II error?

a. Failing to reject the null hypothesis when the null hypothesis is actually false ✓

b. Rejecting the null hypothesis when it is true

c. Selecting a biased sample

d. Misinterpreting the P-value

A Type II error (probability β) is a false negative: failing to detect a real effect when the alternative hypothesis is true.

45 [Challenging] • Target: One-Sample and Two-Sample t-Procedures • Cross-Lesson: 2.01 Sampling Methods & Bias

When estimating a population mean μ with unknown σ, which distribution and test statistic are used?

a. t-distribution with $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ and $df = n - 1$ ✓

b. Standard normal distribution with $z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$

c. Chi-square distribution with $df = n - 1$

d. F-distribution with $df = (n - 1, n - 2)$

When σ is replaced by the sample standard deviation s, the standardized statistic follows Student's t-distribution with $n - 1$ degrees of freedom.

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